d describe the typical behaviour of halogenoalkanes. This will be limited to treatment with:

  i aqueous alkali, e.g. KOH (aq)

  ii alcoholic potassium hydroxide

  iii water containing dissolved silver nitrate

  iv alcoholic ammonia

e carry out the reactions described in 2.10.2d i, ii, iii

f discuss the uses of halogenoalkanes, e.g. as fire retardants and modern refrigerants.

2.11 Mechanisms

Students will be assessed on their ability to:

a classify reactions (including those in Unit 1) as addition, elimination, substitution, oxidation, reduction, hydrolysis or polymerisation

b demonstrate an understanding of the concept of a reaction mechanism and that bond breaking can be homolytic or heterolytic and that the resulting species are either free radicals, electrophiles or nucleophiles

c give definitions of the terms *free radical*, *electrophile* and *nucleophile*

d demonstrate an understanding of why it is helpful to classify reagents

e demonstrate an understanding of the link between bond polarity and the type of reaction mechanism a compound will undergo

f describe the mechanisms of the substitution reactions of halogenoalkanes and recall those in 1.7.2e and 1.7.3e

g demonstrate an understanding of how oxygen, O_2, and ozone, O_3, absorb UV radiation and explain the part played by emission of oxides of nitrogen, from aircraft, in the depletion of the ozone layer, including the free radical mechanism for the reaction and the fact that oxides act as catalysts.